

Dr. José A P M Devienne

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📍 Date of birth / Age: 1st June 1993 / 32

📍 Nationality: Brazilian

📺 José Augusto Devienne

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Education

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|---|-------------|
| • PhD in Geophysics, Peking University, China | 2020 - 2024 |
| • MSc in Applied Physics, São Paulo State University, Brazil | 2018 - 2020 |
| • BSc in Physics, São Paulo State University, Brazil | 2015 - 2017 |

Academic Experiences

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|--|----------------|
| • Research Associate , University of Cambridge, UK | Since Oct/2024 |
| • PhD Student , Peking University, China | 2020 - 2024 |
| • Visiting Researcher , University of Edinburgh, UK | 2022 |
| • MSc Student , São Paulo State University, Brazil | 2018 - 2020 |
| • Intern , São Paulo State University, Brazil | 2016 - 2017 |

Publications

Publications in peer-reviewed journals

3. S. Chen, T. A. Berndt, W. Williams, **J. A. P. M. Devienne**, L. Nagy, P. Wu. (2025). "[Reliable paleomagnetic records from single-vortex iron particles](#)", *Journal of Geophysical Research: Planets*, 130 (8), e2025JE009167.
2. **J. A. P. M. Devienne**, T. A. Berndt, W. Williams, Chen, S. (2025). "[From Disorder to Order: Inheritance of Magnetic Remanence in Tetrataenite-Bearing Meteorites from Multi-Phase Micromagnetic Modeling](#)", *Journal of Geophysical Research: Planets*, 129 (6), 2023JE008268.
1. **J. A. P. M. Devienne**, T. A. Berndt, W. Williams, L. Nagy. (2023). "[Magnetic recording stability of taenite-containing meteorites](#)", *Geophysical Research Letters*, 50, e2022GL102602.

Proceedings

1. **J. A. P. M. Devienne**, T. A. Berndt, W. Williams, Chen, S. (2024) "[From Disorder to Order: Micromagnetic Modeling of Tetrataenite Ordering and Implications for Meteorites Paleomagnetism](#)", *Latinmag Letters, Proceedings Rio de Janeiro, Brazil*, 13, E.02, 1-5.

Preprints

4. **J. A. P. M. Devienne**, Z. Liu, C. Z. Jiang, N. J. Tosca, T. Ginnis, D. Sasselov, R. J. Harrison, S. F. Ozturk. (2026). "[Prebiotic magnetite enables chirality-magnetic surface feedback](#)", arXiv
3. S. Chen, T. A. Berndt, W. Williams, **J. A. P. M. Devienne**, L. Nagy, P. Wu. (2025). "[Reliable paleomagnetic records from single-vortex iron particles](#)", ESS Open Archive Preprint
2. **J. A. P. M. Devienne**, T. A. Berndt, W. Williams, S. Chen. (2023). "[From disorder to order: inheritance of magnetic remanence in tetrataenite-bearing meteorites from multi-phase micromagnetic modeling](#)", ESS Open Archive Preprint
1. **J. A. P. M. Devienne**, T. A. Berndt, W. Williams, L. Nagy. (2023). "[Thermomagnetic recording stability of taenite-containing meteoritic cloudy zones](#)", ESS Open Archive Preprint

Thesis and dissertations

1. **J. A. P. M. Devienne** (2020), "[Acquisition of viscous remanent magnetization in rocks: an experimental approach](#)", MSc degree dissertation
2. **J. A. P. M. Devienne** (2017), "[Método da eletrorresistividade aplicado ao estudo de contaminação de solo no cemitério municipal de Rio Claro, SP](#)", BSc degree dissertation

Non peer-reviewed publications

1. **J. A. P. M. Devienne** (2020), "[Use of social media and Natural Language Processing \(NLP\) in natural hazard research](#)", ArXiv Preprint
2. **J. A. P. M. Devienne** (2020), "[Convolutional neural network for earthquake detection](#)", ArXiv Preprint
3. **J. A. P. M. Devienne** (2020), "[Subduction zone fault slip from seismic noise and GPS data](#)", ArXiv Preprint

Conference and symposium abstracts

1. (Oral) **J. A. P. M. Devienne**, Z. Liu, C. Z. Jiang, N. J. Tosca, T. Ginnis, D. Sasselov, R. J. Harrison, S. F. Ozturk, "Prebiotic magnetite enables chirality-magnetic surface feedback". 19th Biennial Castle Meeting, Collonges-la-Rouge, France, 2026
2. (Oral) **J. A. P. M. Devienne**, Z. Liu, C. Z. Jiang, N. J. Tosca, T. Ginnis, D. Sasselov, R. J. Harrison, S. F. Ozturk, "Prebiotic magnetite enables chirality-magnetic surface feedback". Leverhulme Centre for Life in the Universe (LCLU) Annual Science Day, Cambridge, UK, 2026
3. (Poster) **J. A. P. M. Devienne**, F. Ozturk, Z. Liu, G. Pope, N. Tosca, R. J. Harrison, "You Spin me Round - Exchange-driven reorientation of magnetic states in magnetite by chiral molecules and (potential) implications for the origin of life". Research in Progress Meeting - Mineral Physics Group of the Mineralogical Society of the UK and Ireland, Edinburgh, UK, 2025
4. (Oral) **J. A. P. M. Devienne**, F. Ozturk, Z. Liu, G. Pope, N. Tosca, R. J. Harrison, "You Spin me Round - Exchange-driven reorientation of magnetic states in magnetite by chiral molecules and (potential) implications for the origin of life". Leverhulme Centre for Life in the Universe (LCLU) Annual Science Day, Cambridge, UK, 2025
5. (Oral) **J. A. P. M. Devienne**, F. Ozturk, Z. Liu, G. Pope, N. Tosca, R. J. Harrison, "You Spin me Round - Exchange-driven reorientation of magnetic states in magnetite by chiral molecules and (potential) implications for the origin of life". Magnetic Interactions, Plymouth, UK, 2025
6. (Poster) **J. A. P. M. Devienne**, T. A. Berndt, W. Williams, S. Chen, "Time scales of distinct magnetic records in meteorite cloudy zones". Magnetic Interactions, Plymouth, UK, 2025
7. (Poster) **J. A. P. M. Devienne**, T. A. Berndt, W. Williams, S. Chen, "[Micromagnetic Model of Tetrataenite Ordering in Iron Meteorites: Possibility of Remanence Inheritance From Precursor Taenite?](#)". American Geophysical Union (AGU) Fall Meeting, San Francisco, USA, 2023
8. (Oral) **J. A. P. M. Devienne**, T. A. Berndt, W. Williams, S. Chen, "[From disorder to order: micro-magnetic modeling of tetrataenite ordering and implications for meteorites paleomagnetism](#)". 7th Latinmag, Rio de Janeiro, Brazil, 2023
9. (Poster) **J. A. P. M. Devienne**, T. A. Berndt, W. Williams, L. Nagy, "[Taenite in fast-cooled meteorites thermally stable over billion-year timescales](#)". American Geophysical Union (AGU) Fall Meeting, Chicago, USA, 2022 (Online)
10. (Oral) **J. A. P. M. Devienne**, T. A. Berndt, W. Williams, L. Nagy, "[Taenite in fast-cooled meteorites thermally stable over billion-year timescales](#)". 17th Castle Meeting, Trakošćan, Croatia, 2022
11. (Oral) **J. A. P. M. Devienne**, T. A. Berndt, W. Williams, L. Nagy, "[Taenite in fast-cooled meteorites thermally stable over billion-year timescales](#)". 6th Beijing Earth and Planetary Interior Symposium, Beijing, China, 2022 (Online)

12. (Video) **J. A. P. M. Devienne**, T. A. Berndt, W. Williams, "[Micromagnetic simulations of taenite particles](#)". European Geophysical Union (EGU) General Assmebly 2021 (Online)
13. T. A. Berndt, Chen, H., **J. A. P. M. Devienne**, "[The Bigger, The Better? - Paleomagnetic recording fidelity of weakly interacting clusters of particles, and implications for micro-scale paleomagnetism](#)". European Geophysical Union (EGU) General Assmebly 2021 (Online)
14. (Poster) **J. A. P. M. Devienne**, C. A. Moreira, "[Um estudo teórico sobre o método geofísico da eletrorresistividade](#)". XXIX UNESP Congress of Scientific Initiation, 2017
15. (Poster) **J. A. P. M. Devienne**, C. A. Moreira, "[Um estudo teórico sobre o método geofísico da eletrorresistividade](#)". UNESP Physics Week, 2017
16. (Poster) **J. A. P. M. Devienne**, C. A. Moreira, "Método geofísico da eletrorresistividade aplicado ao estudo de contaminação de solo no cemitério municipal de Rio Claro". UNESP Physics Week, 2016
17. (Poster) **J. A. P. M. Devienne**, E. M. F. Ramos, "O ensino de Física e a curiosidade: estudos de atividades de ensino de eletrostática nos anos iniciais de formação da Educação Básica". XXV UNESP Congress of Scientific Initiation, 2013

Service and Outreach

Reviewing

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| • Geophysical Journal International | Since 2023 |
| • Journal of Geophysical Research: Planets | Since 2024 |

Teaching Experiences

Teaching Assistance

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| • Electromagnetism (BSc in Physics), São Paulo State University | 2017/ 2019 |
| Teaching assistant for undergraduate level course on Electromagnetism | |

Awards and Grants

Awards

- Honorable mention to oral presentation in the 7th LatinMag 2023: "[From disorder to order: micromagnetic modeling of tetrataenite ordering and implications for meteorites paleomagnetism](#)".

Grants

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| • Institute for Rock Magnetism (IRM) Visiting Researcher Fellowship (US\$1200.00) | 2025 |
| • National Natural Science Foundation of China (Co-I) (590.000 RMB) | 2021-2024 |
| • Chinese Government Scholarship (PhD Scholarship) (158.400 RMB) | 2020-2024 |
| • CAPES MSc Scholarship (R\$33.000) | 2018-2020 |
| • CAPES Scientific Initiation Scholarship (R\$2800.00) | 2016-2017 |
| • BAAE scholarship (R\$8400.00) | 2015-2016 |

Extracurricular Courses

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| • Machine Learning Engineering (360 hours)
FIAP, Brazil | 2025 |
| • Computational Material Physics (60 hours)
Ghent University, Belgium | 2022 |

• Advanced School in Big Data Analysis (8 hours) University of São Paulo, Brazil	2021
• The Geomagnetic Field and The Core Dynamics (20 hours) University of São Paulo, Brazil	2018
• Introduction to Seismology (20 hours) University of São Paulo, Brazil	2017
• Seismic Methods Applied to The Investigation of Shallow Water Environments (20 hours) University of São Paulo, Brazil	2017
• Introduction to Geophysics (9 hours) State University of Campinas, Brazil	2017
• Seismic Diffraction: Introduction and Examples (9 hours) State University of Campinas, Brazil	2017

Technical Skills

- **Programming Languages & Frameworks** Python
- **Writing softwares** Latex, MS Office, Adobe Acrobat, LibreOffice
- **Scientific Computing** MATLAB & GNU Octave, Mathematica, MERRILL, Quantum Espresso, VESTA, Anaconda, Igor Pro
- **Parallel Computing** MPI, OpenMP
- **Data Processing** Excel, Pandas
- **Machine Learning** Scikit-learn, Keras, TensorFlow
- **Engineering Software** ParaView, Cubit
- **Experimental methods** Quantum Diamond Microscope (QDM), Superconducting quantum interference device (SQUID), spinner (JR6), alternating gradient (AGM) and vibrating sample (VSM) magnetometers; inductor of isothermal and anhysteretic remanent magnetizations (Agico LDA5); furnace (TD48) and alternating field demagnetizing devices (LDA5); scanning electron microscope (SEM); X-ray diffractometry (XRD); electrical tomography (ABEM Terrameter LS 2). Synchrotron methods (e.g., ptychography).
- **Field Work** 3-days field work for collecting geoelectrical tomographic data from subsurface of a public cemetery located in Rio Claro, SP (Brazil); 1-day field work in Piracicaba (Brazil) for collecting rock samples for viscous magnetization experiments.